Xiaochuan Li

M.Sc. Human and AI · University of Technology Nuremberg

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EDUCATION

M.Sc., Human and AI 2025 — 2027

University of Technology Nuremberg (UTN)

Focus: causal interpretability and its intersection with language — when model behavior is licensed by the structure that should support it, and how that structure differs across languages.

German language program 2024 — 2025

University of Marburg · intensive German for graduate study

B.A., Philosophy 2020 — 2024

Shenzhen University · College of Humanities

Senior thesis: *On Leibniz's Logical Theory* (advisor: Zang Yong)

Coursework: logic, history of philosophy, analytic philosophy, mathematical logic; electives in cognitive psychology and linguistics.

RESEARCH

Hierarchical Causal Evidence

Core research program

A sequence from binary grounding certification, to behavioral path localization, to quantitative grading of causal structure use.

Latent Control States and Cross-lingual ProtoBias

Mechanistic and applied extensions

Extensions of the core causal-licensing question into activation-space states and multilingual/multimodal prototype shortcuts.

Typological Grounding

Emerging direction

Carrying the structure-versus-surface question across languages that encode meaning very differently — through morphology, word order, or context.

PAPERS IN PROGRESS

Stable Is Not Grounded. Paper 1: binary certification for whether the licensing structure is load-bearing.

Isotrace. Paper 2: behavioral path localization via path-encoded fictional names.

Causal Inferential Yield (CIY). Paper 3: quantitative grading over certified or controlled causal-structure use.

EXPERIENCE

IT Student Assistant Apr 2026 — present

University of Technology Nuremberg

Hardware maintenance and general IT support for staff and teaching labs.

SKILLS

Working knowledge	Python (NumPy, pandas, scikit-learn), LaTeX, Git
Currently learning	PyTorch & HuggingFace transformers (incl. LoRA fine-tuning), interventional causal-inference tooling (do-interventions, causal graphs)

LANGUAGES

Mandarin Chinese (native) · German (C1) · English (C1)

INTERESTS

Causal inference and its application to interpretability — increasingly at its intersection with language: how typologically different languages encode meaning explicitly or implicitly, and whether models represent that structure or a surface proxy. The boundary between behavioral and mechanistic methods. Philosophy of measurement and of language in ML. World-model approaches (LeCun, Pearl) and the limits of pure-LLM paradigms.